

PROJECT BLUEPRINT: SCALABLE HOME ECOSYSTEM

Phase 1: ESP32 Integration & Intelligence Layer

1. Executive Summary

This project aims to build a modular, local-first home automation system using **Home Assistant** as the orchestrator and **ESP32** microcontrollers as the hardware interface. The system will bridge physical sensors/actuators with cloud data (weather) and local AI (LLM) for intelligent decision-making.

2. Architecture Overview

The Workflow:
ESP32 (Sensors/Actuators) → WiFi (MQTT/ESPHome) → Home Assistant Server → OpenWeatherMap API → Local LLM (Ollama/LocalAI) → Dashboard (PC Control).

3. Tool Comparison & Selection

3.1 Connectivity Protocol

Protocol	Pros	Cons	Selection
HTTP / REST	Simple to test	High overhead, slow	-
MQTT	Lightweight, standard	Requires a Broker	Recommended for generic devices
ESPHome	Native HA integration	Tied to HA ecosystem	SELECTED (Best for ESP32)

3.2 Local LLM Integration

Tool	Hardware Req.	Complexity	Selection
OpenAI API	None (Cloud)	Low	- (Not Local)
Ollama	Moderate (8GB+ RAM)	Low (One-click)	SELECTED
LocalAI	Moderate	High	Alternative

4. Bill of Materials (BOM)

- **Controller:** ESP32 DevKit V1 (Built-in WiFi/BT)
- **Actuators:** 5V LED, SG90 Servo Motor, 5V Relay (for Switch)
- **Sensors:** DHT22 (Temperature & Humidity)
- **Host:** Raspberry Pi 4 (4GB+) or an old PC/Mini-PC (NUC) to run Home Assistant.

5. Requirement Specifications

Functional Requirements

- **Real-time Monitoring:** Dashboard must show live temperature and LED/Servo states.
- **Remote Control:** Toggle LED and move Servo slider via PC interface.
- **Logic Engine:** Compare local sensor temp vs. City Weather API; if difference > 5°C, send suggestion via LLM.
- **Logging:** DB-backed history for all sensor readings and command logs.

Non-Functional Requirements

- **Scalability:** ESPHome allows adding new ESP32 nodes without writing C++ code for each.
- **Security:** All LLM processing remains on the local network.

6. Implementation Timeline (4-Week Sprint)

Phase	Tasks	Duration
Week 1: Infrastructure	Install Home Assistant OS (HAOS), setup MQTT Broker, Install Ollama.	5-7 Days

Phase	Tasks	Duration
Week 2: Hardware Node	Flash ESP32 with ESPHome, wire DHT22, LED, and Servo. Calibrate sensors.	5 Days
Week 3: Intelligence	Integrate Weather API, setup Home Assistant Automations, link Ollama API.	6 Days
Week 4: Frontend & Logs	Design Lovelace Dashboard (UI), configure History/Logbook views. Final testing.	4 Days